



Agenda this week

Last week - project Review

Bootstrap



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KdG





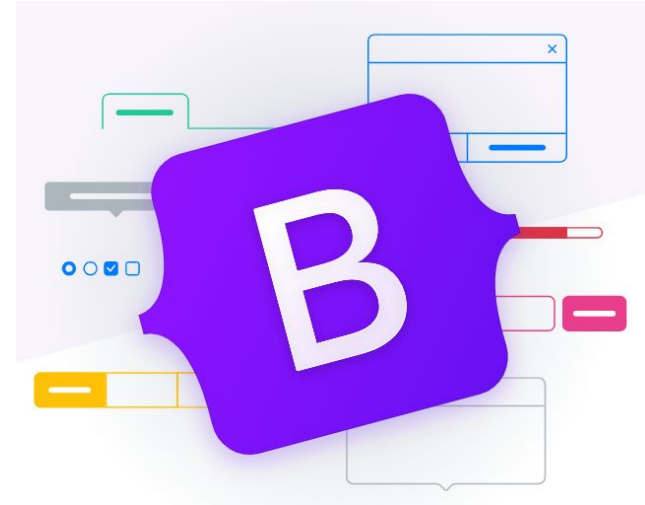
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Bootstrap

Bootstrap

- Bootstrap is a CSS framework to jumpstart web development providing
 - [A responsive grid system](#)
 - Prebuilt components with behaviour built-in using javascript
 - [Navbar](#)
 - [Form](#)
 - [Carousel](#)
 - ...



Bootstrap: add dependency using WebJars

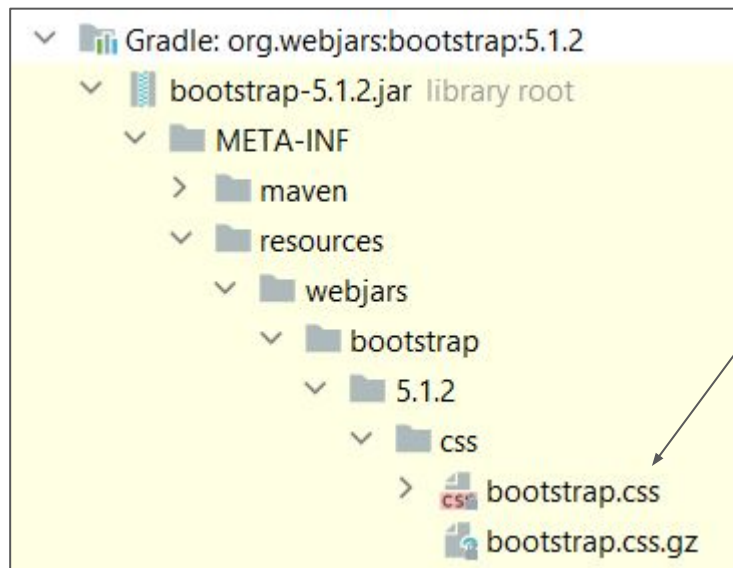
- We can use Gradle to download Client side web libraries like Bootstrap for us!
- [WebJars](#) wraps them into a jar (Java Archive) that can be added as a dependency to the `build.gradle` file for your server

```
implementation("org.webjars:bootstrap:5.3.8")
```

WebJars Documentation			
Popular WebJars			
Name	Versions	Build Tool: SBT / Play 2 Maven Ivy Grails Gradle Bazel Leiningen	
Swagger UI	5.7.2	+	compile "org.webjars.swagger-ui:5.7.2"
jQuery	3.7.1	+	compile "org.webjars.jquery:3.7.1"
npm	5.0.0-2	+	compile "org.webjars.npm:5.0.0-2"
Bootstrap	5.3.2	+	compile "org.webjars.bootstrap:5.3.2"
Font Awesome	6.4.2	+	compile "org.webjars.font-awesome:6.4.2"
jQuery	3.7.1	+	compile "org.webjars.jquery:3.7.1"
viz.js-graphviz.js	2.1.3	+	compile "org.webjars.viz.js:graphviz:2.1.3"
Popper.js	2.11.7	+	compile "org.webjars.popper.js:2.11.7"
balanced-match	1.0.2	+	compile "org.webjars.npm:balanced-match:1.0.2"
Data Tables	1.13.5	+	compile "org.webjars.datatables:1.13.5"
@angular/http	6.0.0-beta.10	+	compile "org.webjars.npm:angular_1100-6.0.0-beta.10"
jQuery UI	1.13.2	+	compile "org.webjars.jquery-ui:1.13.2"
node.js	9.11.2	+	compile "org.webjars.node.js:9.11.2"



WebJars : check the External Libraries in IntelliJ

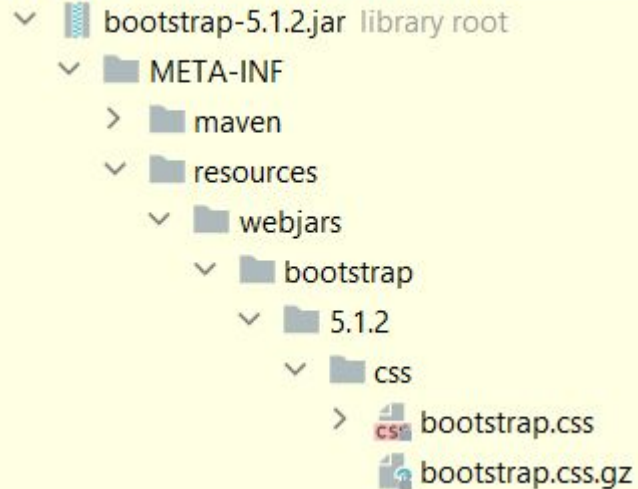


The Bootstrap jar is included in the classpath, so you can access the bootstrap CSS and JavaScript files from within Thymeleaf...
Adapt for your Bootstrap version :(

```
<link rel="stylesheet" th:href="@{/webjars/bootstrap/51.2/css/bootstrap.min.css}"/>
<script defer th:src="@{/webjars/bootstrap/51.2/js/bootstrap.bundle.min.js}">
</script>
```

Webjars-locator: version agnostic references!

```
implementation("org.webjars:bootstrap:5.3.8")  
implementation("org.webjars:webjars-locator-core:0.59")
```



If we add the webjars-locator dependency to our project, Spring Boot will use this library to deduce the correct version of the webjar: we don't need to add the version number in our references in HTML!

```
<link rel="stylesheet" th:href="@{/webjars/bootstrap/css/bootstrap.min.css}"/>  
<script defer th:src="@{/webjars/bootstrap/js/bootstrap.bundle.min.js}">  
</script>
```


Bootstrap: does it work?

Let's add a Bootstrap alert to our index.html

```
<div class="container ">
  <div class="my-3 alert alert-warning alert-dismissible" role="alert">
    <span class="me-1 badge text-bg-warning">Dogs beware!</span>
    I am a dog that is 3 years or older.
    <button type="button" class="btn btn-close" data-bs-dismiss="alert">
    </button>
  </div>
  <h3><a th:href="@{/doghotel}"> Enter our dog hotel</a></h3>
</div>
```

Add bootstrap CSS classes to style your HTML

Bootstrap uses attributes starting with **data-bs-** to hook up javascript. Clicking the X should close the Alert!

Dogs beware! I am a dog that is 3 years or older.



Bootstrap: why the script tag?

```
<link rel="stylesheet" th:href="@{/webjars/bootstrap/css/bootstrap.min.css}"/>  
<script defer th:src="@{/webjars/bootstrap/js/bootstrap.bundle.min.js}">  
</script>
```

Bootstrap is more than just CSS: it also includes some JavaScript code, for example to close the Alert when clicking on the X!

Dogs beware! I am a dog that is 3 years or older.



Bootstrap: what's this .min.css and .bundle.min.js?

```
<link rel="stylesheet" th:href="@{/webjars/bootstrap/css/bootstrap.min.css}"/>  
<script defer th:src="@{/webjars/bootstrap/js/bootstrap.bundle.min.js}">  
</script>
```

.min stands for “minified”: it is CSS but all unnecessary characters are removed to make it as small as possible.

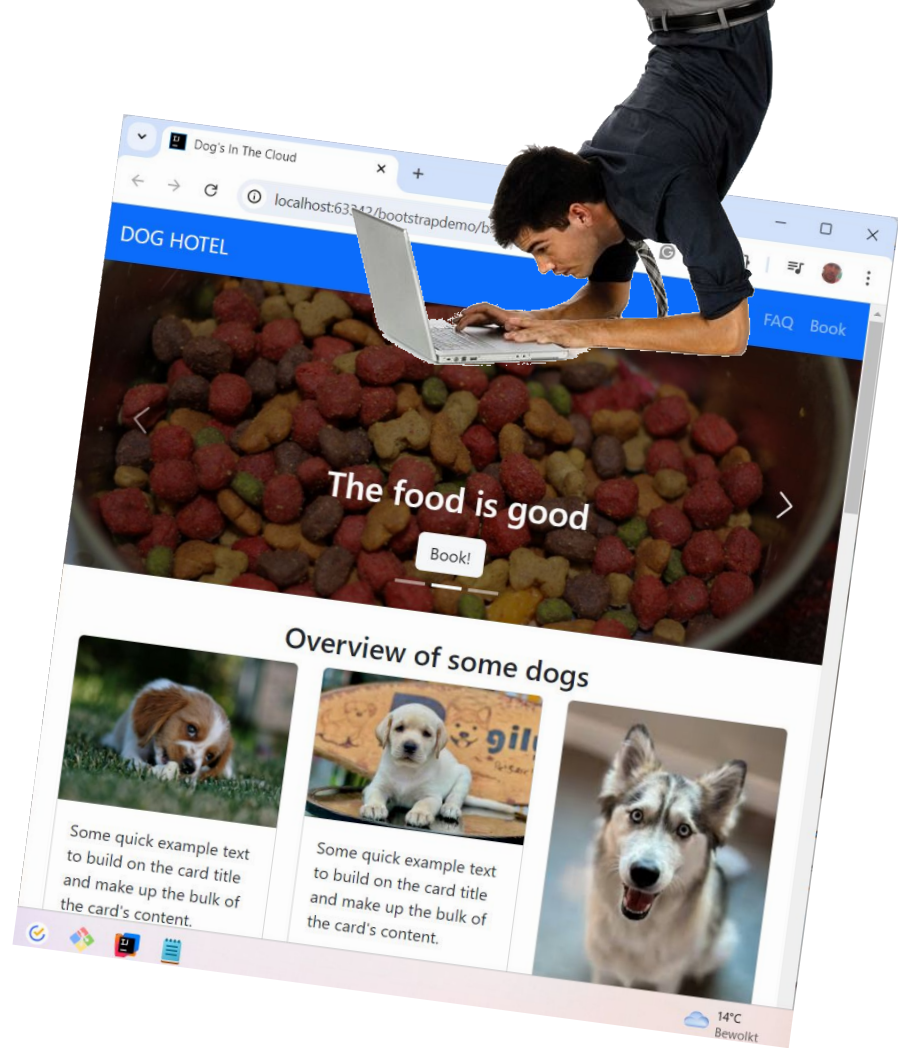
The **bundle** means that it bundles more than one js file: it also includes the popper.js library which Bootstrap uses for popups...

Dogs beware! I am a dog that is 3 years or older.



Demo

- Dog Hotel demo: adding Bootstrap



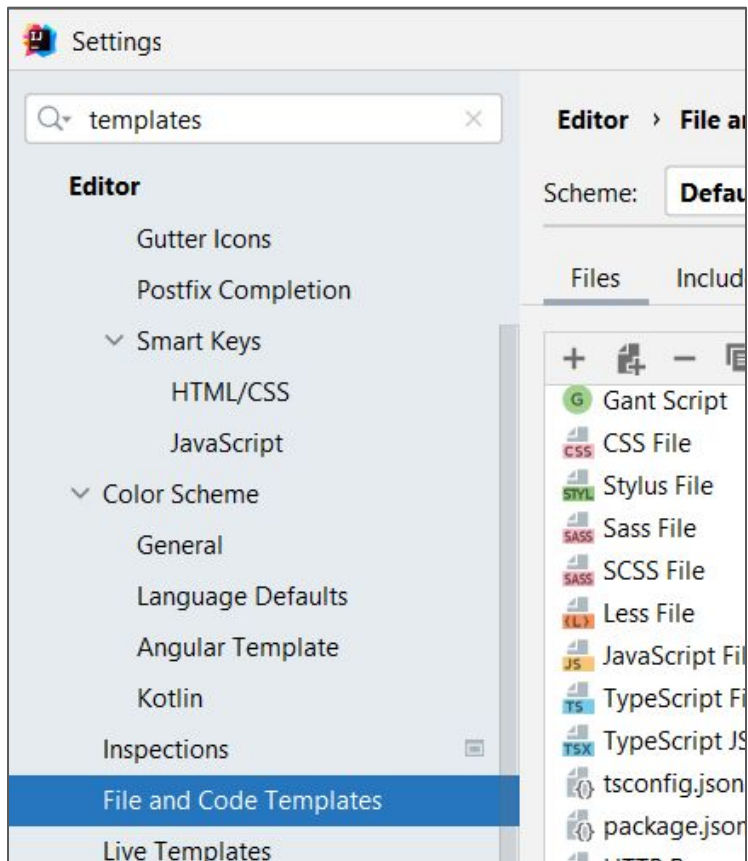
Exercise 1: Bootstrap with webjars

- Use the SMSWithThymeleaf project
- Add dependencies to the Bootstrap webjar and webjars-locator
- Include the `<link>` and `<script>` tags: does it work?
- What happens if you remove the script tag?
- Tip: add a new template to IntelliJ that create a Thymeleaf starter page which includes the correct `<link>` and `<script>` tag



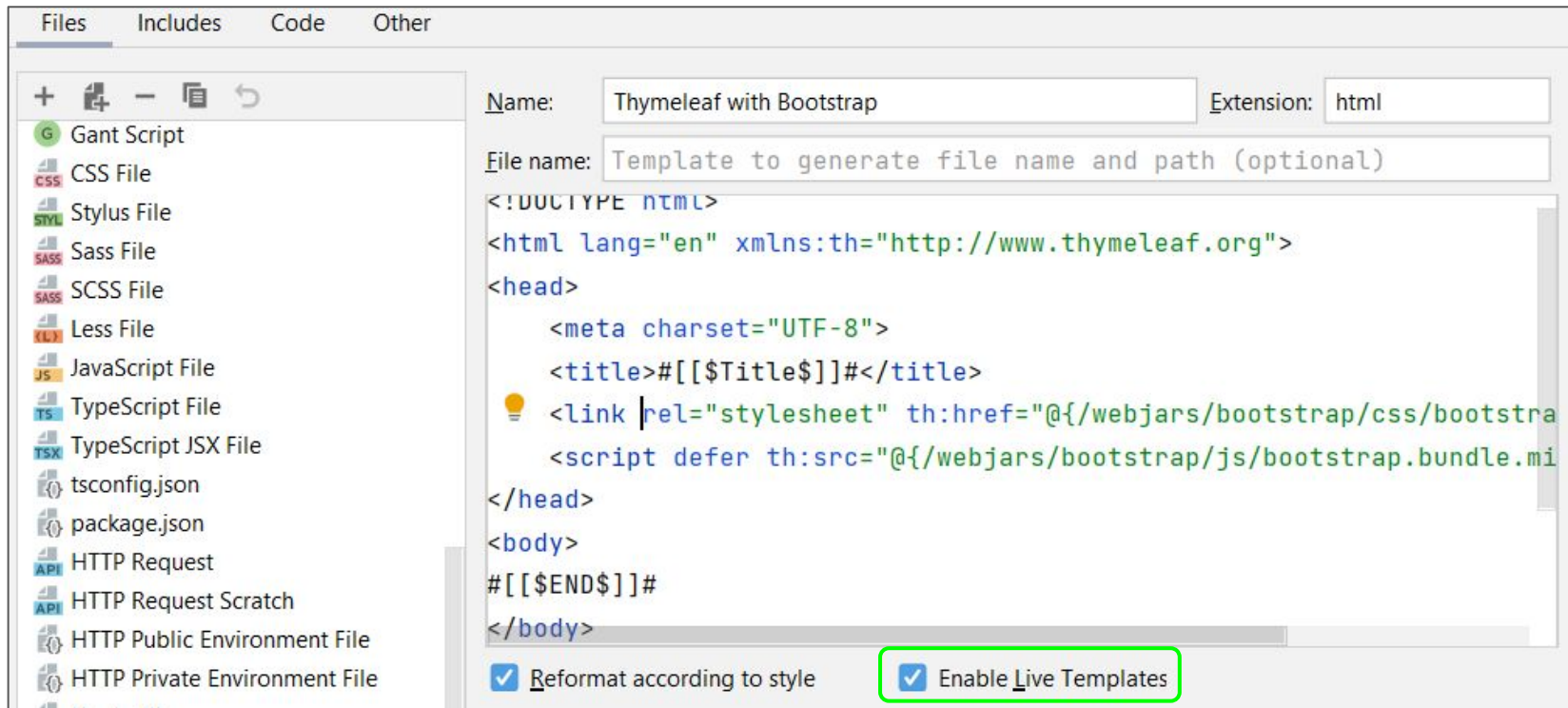


Tip: add a Code Template to IntelliJ!



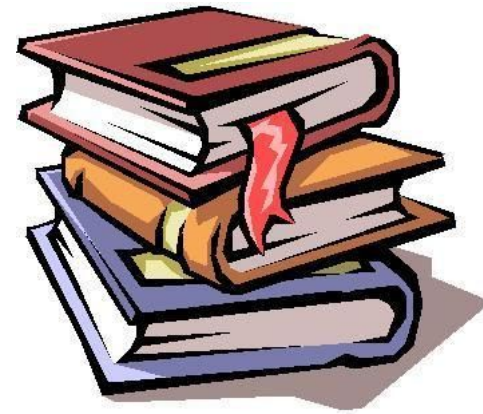


Tip: add a Code Template to IntelliJ!



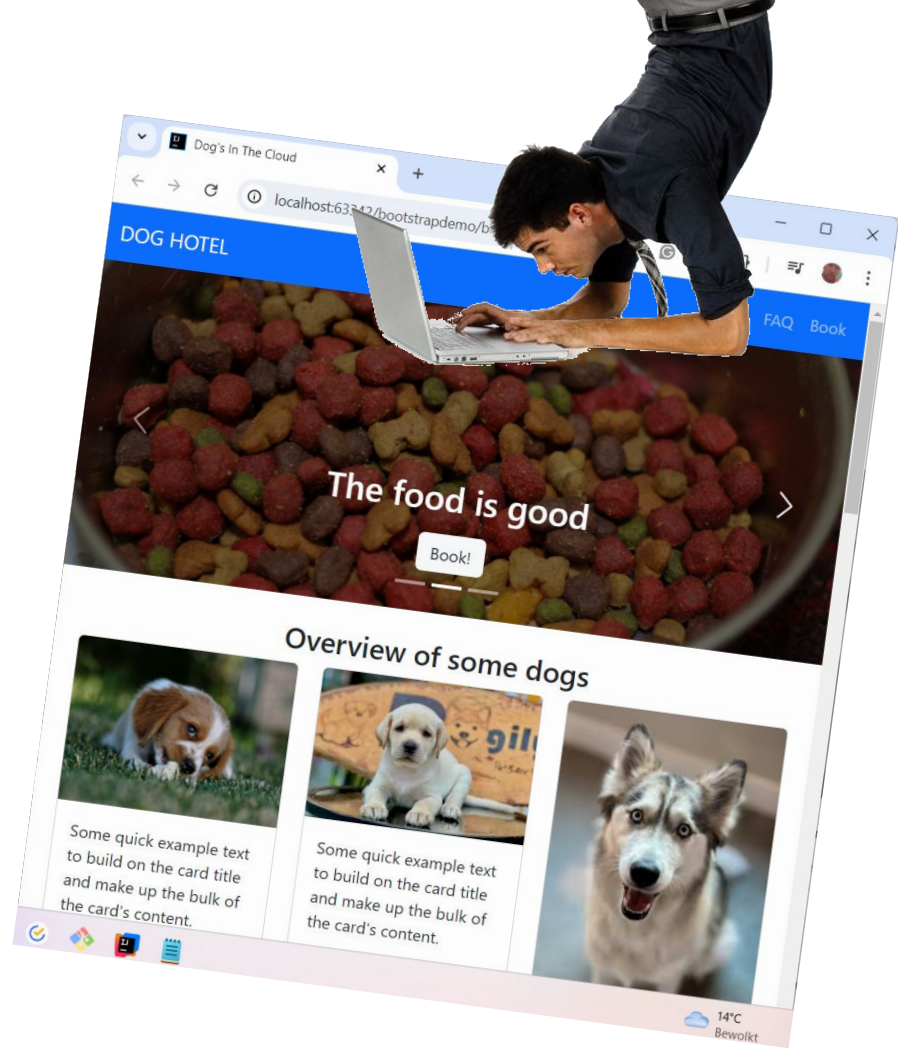
Bootstrap tutorial

- [The Bootstrap documentation](#) is good with live examples embedded.
- It covers a lot of topics, you have to master!
 - Layout
 - Containers
 - Grid System
 - Fixed and Fluid Layouts
 - Responsive Layouts
 - Content
 - Typography
 - Tables
 - Buttons
 - [Bootstrap icons](#)
 - ...
 - There is also a [section with more examples](#).
- There is a good tutorial at <https://www.tutorialrepublic.com/twitter-bootstrap-tutorial>



Demo

- Dog Hotel demo:
 - Bootstrap Components:
 - Navbar - Carousel - Cards - ...



Exercise 2: Add navigation elements

- Check the navbar section in [The Bootstrap documentation](#)
- Add a Bootstrap navigation element to the top of the allstudents page
 - Home - Add Student - About
- Use thymeleaf fragments!

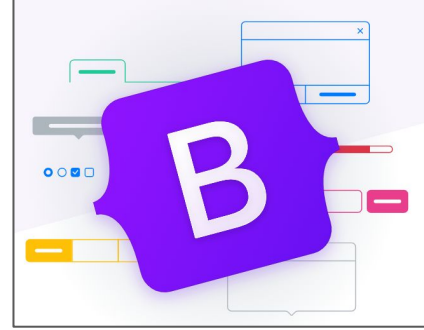


The Bootstrap Grid System: basic idea

takes 3/12 of row

takes 4/12 of row

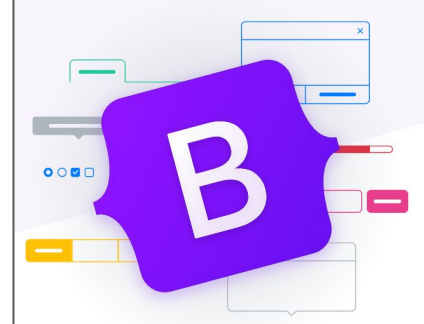
takes 5/12 of row



```
<div class="container">
  <div class="row">
    <div class="col-3">
      takes 3/12 of row
    </div>
    <div class="col-4">
      takes 4/12 of row
    </div>
    <div class="col-5">
      takes 5/12 of row
    </div>
  </div>
</div>
```

- All content is in a `.container`
- Containers have horizontal `.row` children
- The immediate children of `.row` elements are vertical `.col` (column) elements
- A row is divided into 12 equal columns. You can define the number of columns a child occupies by adding a number to the class: for example `.col-3` use 3 out of the 12 columns.
- `.col` without a number divides remaining columns evenly

Responsive Grid Breakpoints



	xs <576px	sm ≥576px	md ≥768px	lg ≥992px	xl ≥1200px	xxl ≥1400px
Container <i>max-width</i>	None (auto)	540px	720px	960px	1140px	1320px
Class prefix	<i>.col-</i>	<i>.col-sm-</i>	<i>.col-md-</i>	<i>.col-lg-</i>	<i>.col-xl-</i>	<i>.col-xxl-</i>
# of columns	12					
Gutter width	1.5rem (.75rem on left and right)					
Custom gutters	Yes					
Nestable	Yes					
Column ordering	Yes					

The Bootstrap Grid System: combining .col

col-sm-3 col-md
col-sm-9 col-md-8
col-sm-12 col-md

col-sm-3 col-md
col-sm-9 col-md-8
col-sm-12 col-md

col-sm-3 col-md
col-sm-9 col-md-8
col-sm-12 col-md

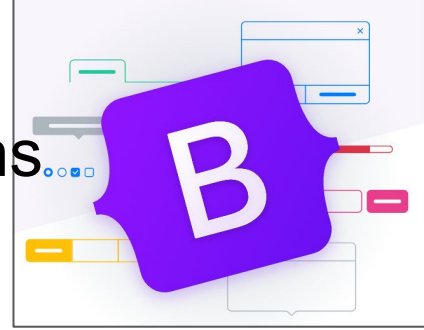
combine bootstrap classes with different breakpoints for responsive pages

- no breakpoint: default each div takes entire width (3 rows)
- sm: above sm 3-9-12 ratio (2 rows)
- md: above md 2-8-2 ratio (1 row)

```
<div class="row">  
  <div class="col-sm-3 col-md">col-sm-3 col-md</div>  
  <div class="col-sm-9 col-md-8">col-sm-9 col-md-8</div>  
  <div class="col-sm-12 col-md">col-sm-12 col-md</div>  
</div>
```



The Bootstrap Grid System: equal width columns



Row with equal # of children

Still the same

Still the same

Still the same

Still the same

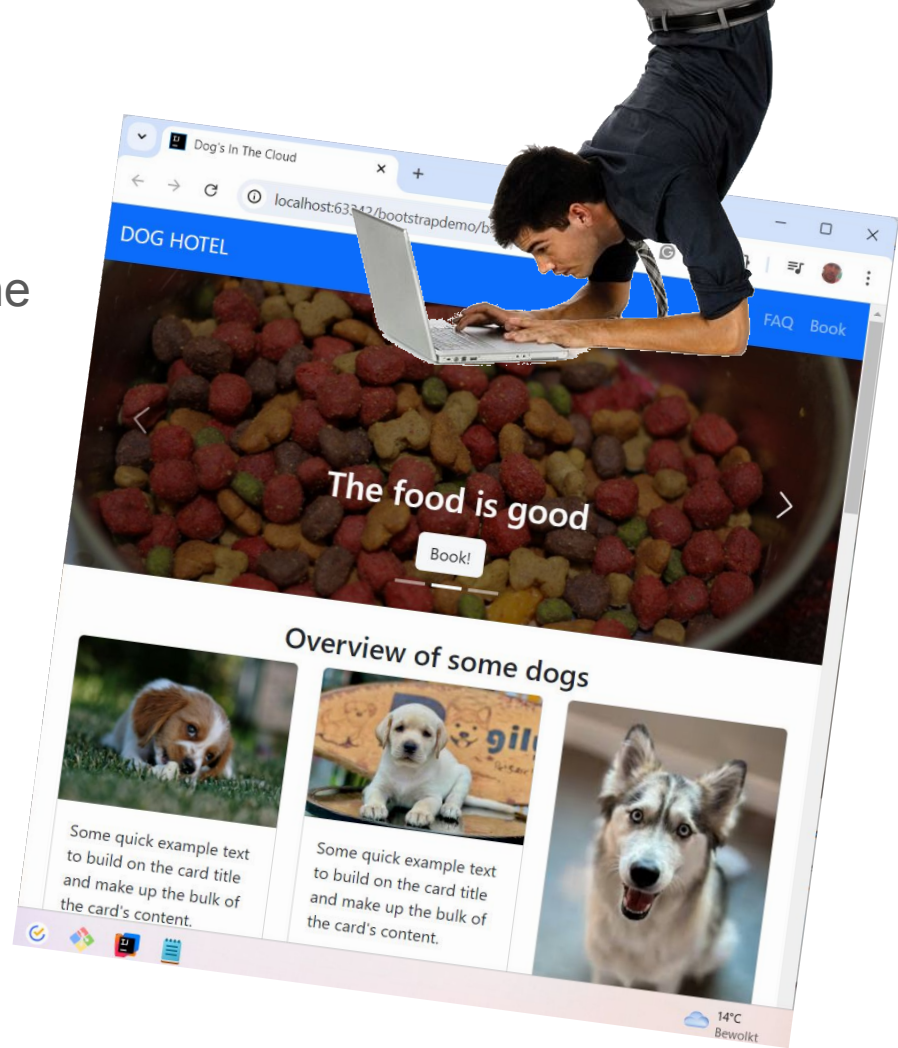
- if all cols have the same width you can use *row-cols-breakpoint-number* on the **row**
- number is the number of children on a row (5 on large, 2 on medium)

```
<div class="container">
  row with equal # of children
  <div class="row row-cols-lg-5 row-cols-md-2">
    <div class="col">Still the same</div>
    <div class="col">Still the same</div>
    <div class="col">Still the same</div>
    <div class="col">Still the same</div>
    <div class="col">Still the same</div>
  </div>
</div>
```

<https://codepen.io/jdrijke/pen/MYWaxmq>

Demo

- In the Dog Hotel: putting the cards in the grid system...



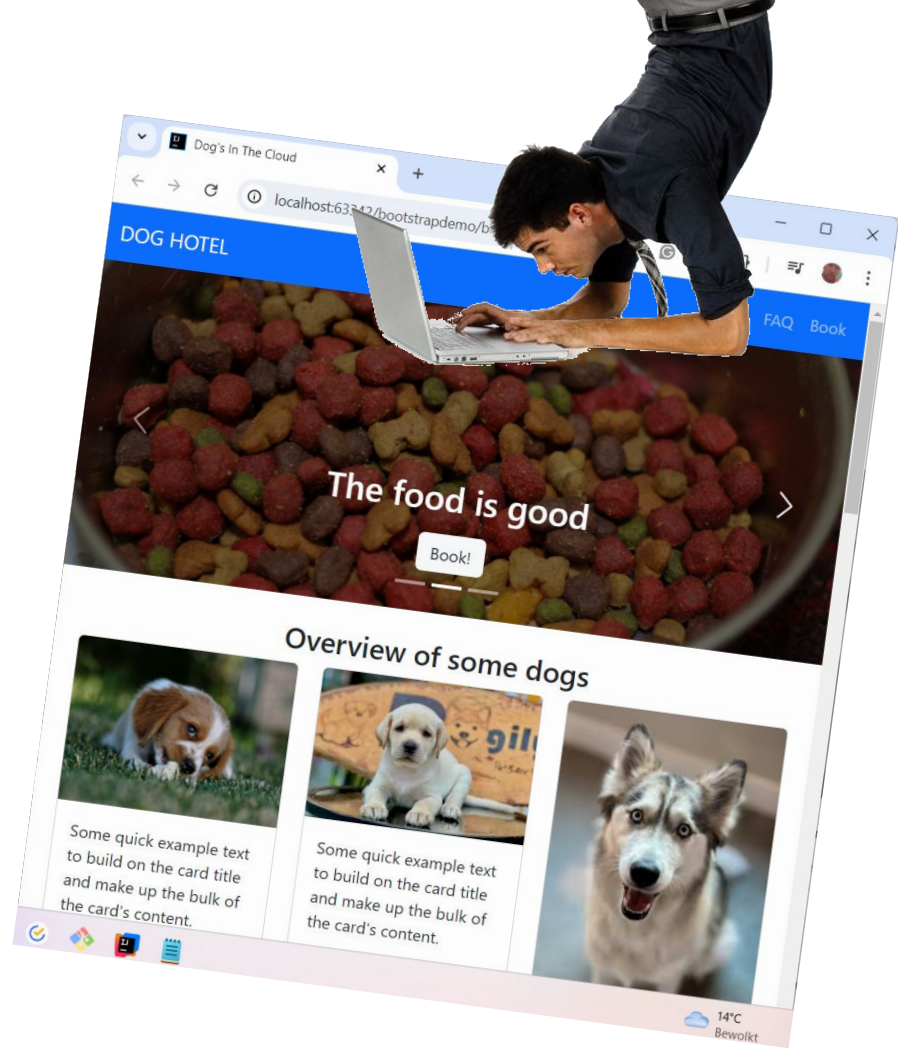
Exercise 3: Bootstrap gridsystem & forms

- Read the tutorial (or the bootstrap doc)
 - <https://www.tutorialrepublic.com/twitter-bootstrap-tutorial/bootstrap-grid-system.php>
 - <https://www.tutorialrepublic.com/twitter-bootstrap-tutorial/bootstrap-forms.php>
 - form demo: <https://codepen.io/jdrijke/pen/GLVMLK>
- Add a addstudent page to the SMSWithThymeleaf application
 - Create a nice looking form to be able to add a student
 - Be sure to put the same footer and navbar on the page (use Fragments!)
 - Add some extra fields to the student:
 - Adres
 - Phone
 - E-mail
 - Test on different screen sizes!



Demo

- In the Dog Hotel: layout of a table



Exercise 4: Bootstrap tables

- Read the section in the tutorial (or the bootstrap doc)
- <https://www.tutorialrepublic.com/twitter-bootstrap-tutorial/bootstrap-tables.php>
- Update the layout of your table in the allstudents page. Use bootstrap to do the coloring of the odd rows...



Bootstrap icons

- Bootstrap icon characters are in a separate [npm package](#)
 - Any npm package can be added to gradle as a webjar. In build.gradle.kts:

`implementation("org.webjars.npm:bootstrap-icons:1.11.3")`
- In your webpage load the CSS from gradle In your webpage load the CSS from gradle
 - add it to your bootstrap-thymeleaf template!

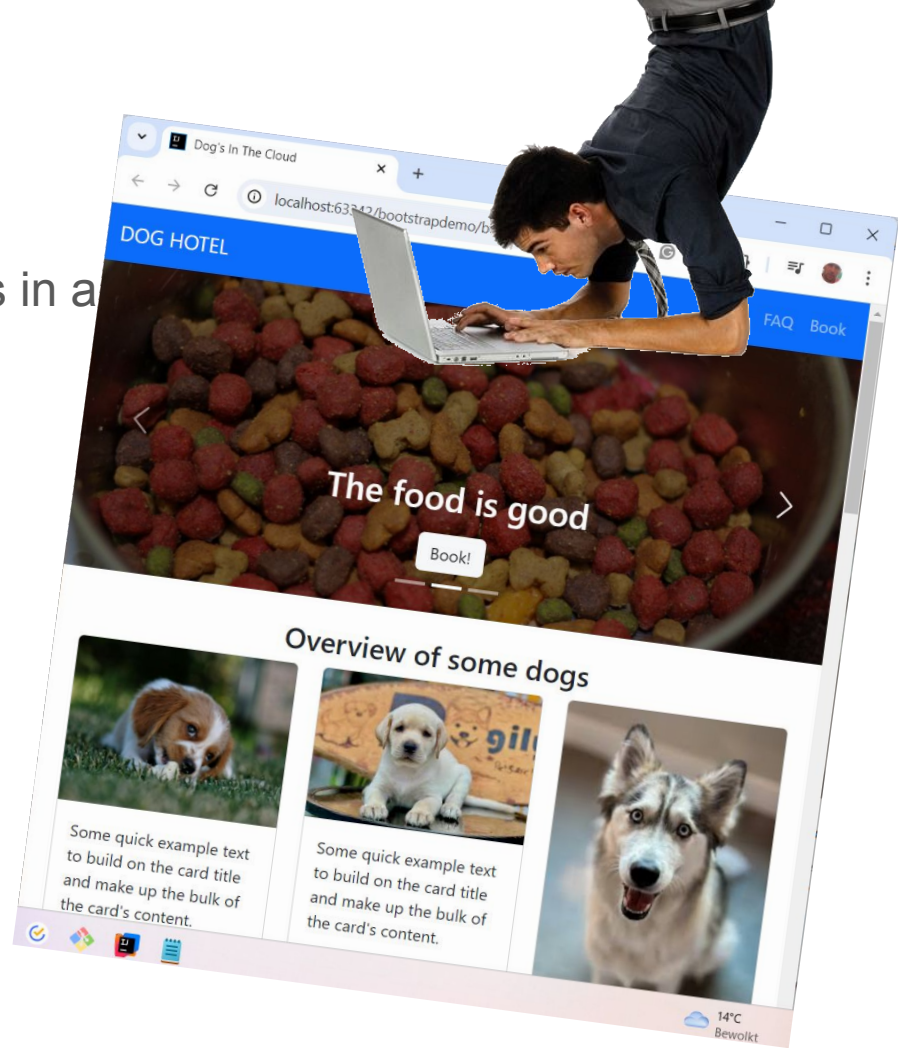
```
<link rel="stylesheet" th:href="@{/webjars/bootstrap-icons/font/bootstrap-icons.css}">
```

- Note: instead of hosting web dependencies on your server using webjars, you can also load bootstrap and bootstrap icons from a CDN on the internet:

```
<link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.3/font/bootstrap-icons.css">
```

Demo

- In the Dog Hotel: using Bootstrap Icons in a footer



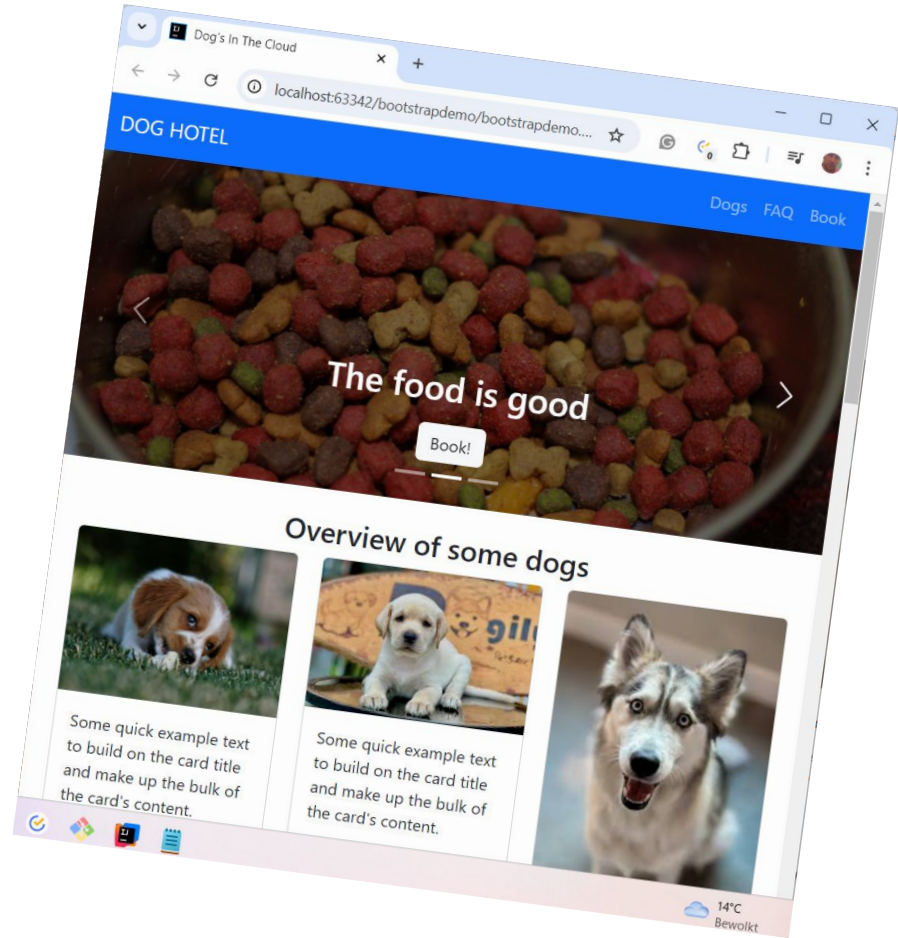
Exercise 5: Bootstrap icons

- Read the section in the tutorial
- <https://www.tutorialrepublic.com/twitter-bootstrap-tutorial/bootstrap-icons.php>
- Lookup the different Bootstrap icons and try adding some to your pages:
 - Add a special icon to students born after 2000?
 - Add an icon to your footer information?
 - ...



There is more...

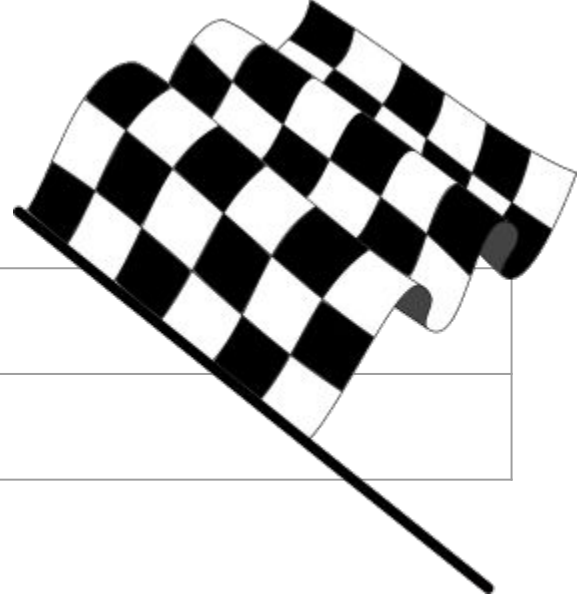
- [The Bootstrap documentation](#)
- Utility classes for margin, padding, font-size, display, flex...
- Custom CSS?
- ...



Agenda this week

Last week - project Review

Bootstrap



Project

- Use Thymeleaf and Bootstrap to work out the Views of your web application:
 - Include Bootstrap using webjars
 - In your pages:
 - show the list of main entities as bootstrap cards with a name, an attribute and an image
 - add the name of an image file to your main entity
 - store image files corresponding to those names somewhere under resources/static
 - Make your pages responsive: they look nice on any screen size
 - your image cards should display in 1, 2 and 4 columns at different screen widths
 - If I click on an item, I get a new page containing all details of this item.
 - Have an element in a page that only shows at large displays (hint: d-none)
 - Add a bootstrap Navbar (use fragments!), add a footer (use fragments!)
 - Use bootstrap forms to update the look of your addXxx pages.
 - We appreciate a personal touch to your site that we don't find in other submissions.
 - Take a look at bootstrap components/utilities/icons... to see if there is an element you can use to make your site stand out
 - favour Bootstrap styling over custom CSS
 - Implement client-side validation (required fields)

